Foundations of Modern Financial Risk

What You Need to Know Before Writing Code

Risk Analyst Project

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Abstract

A concise, example-driven introduction to the core concepts of modern financial risk analysis. Each idea is motivated by a concrete scenario, then generalized. We assume familiarity with probability, statistics, and basic stochastic processes; the focus is on *how these tools are used in risk management* and *why the field evolved the way it did*. References point to deeper material for each topic.

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1 What Risk Means in Finance

Risk in finance is not synonymous with volatility. It is the *possibility of loss* — specifically, loss that exceeds what you planned for.

Example 1.1 (A simple portfolio)**.** You hold \$100,000 split equally between the S&P 500 (SPY) and 20-year Treasury bonds (TLT). On a typical day the portfolio moves ±\$500. But on March 16, 2020, the S&P dropped 12% in a single session while Treasuries also sold off. Your portfolio lost \$8,200 — sixteen times the "normal" day.

Risk management answers: *How often can losses this large happen? How much capital should I reserve? What if it gets even worse?*

The field has converged on three interlocking concerns:

Measure Quantify the loss distribution: how bad can it get, and with what probability?

Manage Set limits, allocate capital, hedge, diversify.

Monitor Backtest your models against reality; adjust when they fail.

2 Loss Distributions and Tail Behavior

2.1 From Returns to Losses

Let $r_t = \ln(P_t/P_{t-1})$ be the log-return of an asset. For a portfolio with weights $\boldsymbol{w}$ and asset returns $\boldsymbol{r}_t$, the portfolio return is $r_p = \boldsymbol{w}^\top \boldsymbol{r}_t$. The **loss** is $L = -r_p$ (we flip the sign so that losses are positive).

The central object in risk is the **loss distribution** $F_L(x) = \mathbb{P}(L \leq x)$.

2.2 Why the Normal Distribution Fails

Example 2.1 (Fat tails in practice)**.** Daily S&P 500 returns from 1990–2025 have:

- Kurtosis ≈ 11.5 (normal $= 3$). Tails are *much* heavier.
- The Oct 19, 1987 crash (-22.6%) is a $\sim$25-sigma event under normality — probability $\approx 10^{-138}$.
- In reality, moves $> 4\sigma$ happen about 10x more often than the normal predicts.

Key Insight

Financial returns exhibit **heavy tails** (excess kurtosis), **volatility clustering** (large moves follow large moves), and **asymmetry** (crashes are sharper than rallies). Any risk model that ignores these features will systematically underestimate losses.

Better distributional choices:

- **Student-t**: $f(x;\nu) \propto (1 + x^2/\nu)^{-(\nu+1)/2}$. Tails decay polynomially, not exponentially. Typical fit: $\nu \approx 4$–6 for equity returns.
- **Generalized Pareto** (for extreme tails only): the correct asymptotic model for tail exceedances (see Section 4).
- **Skewed-t** (Hansen, 1994): adds asymmetry to Student-t. Used as the innovation distribution in GARCH models.

Go Deeper

- Cont, R. (2001). "Empirical properties of asset returns: stylized facts and statistical issues." *Quantitative Finance*, 1(2), 223–236. *The* survey of return properties.
- McNeil, Frey & Embrechts (2015). *Quantitative Risk Management*, Ch. 3. Distributional models for returns.

3 Risk Measures: VaR, Expected Shortfall, and Coherence

3.1 Value-at-Risk

Definition 3.1 (Value-at-Risk)**.** For a loss L with distribution F_L and confidence level $\alpha \in (0, 1)$:

$$\mathrm{VaR}_\alpha(L) = F_L^{-1}(\alpha) = \inf\{x : F_L(x) \geq \alpha\}$$

Example 3.1 (Computing VaR)**.** Your SPY/TLT portfolio has 252 daily returns over the past year. Sort the losses from smallest to largest. At $\alpha = 0.95$ (95%), $\mathrm{VaR}_{0.95}$ is the 13th-worst loss (since $252 \times 0.05 \approx 13$).

Suppose that value is \$1,850. Interpretation: "On 95% of days, losses do not exceed \$1,850." Equivalently: "We expect to lose more than \$1,850 about once every 20 trading days."

Three ways to compute VaR:

1. **Historical simulation**: sort historical losses, read off the quantile.
2. **Parametric (variance-covariance)**: assume $L \sim N(\mu, \sigma^2)$, then $\mathrm{VaR}_\alpha = \mu + \sigma\,\Phi^{-1}(\alpha)$. Fast but relies on normality.
3. **Monte Carlo**: simulate thousands of portfolio paths, compute losses, read off the quantile. Most flexible; handles non-linear instruments.

3.2 The Problem with VaR

Warning

VaR tells you the *threshold* but nothing about *how bad it gets beyond that threshold.* Two portfolios can have identical VaR but very different tail losses.

Example 3.2 (VaR blindness)**.** Portfolio A loses exactly \$2,000 on the 5% worst days. Portfolio B loses \$2,000 on most of those days, but occasionally loses \$50,000. Both have $\mathrm{VaR}_{0.95} = \$2{,}000$. A risk manager using only VaR would treat them identically.

Worse: VaR is **not subadditive** in general. It is possible that $\mathrm{VaR}(A + B) > \mathrm{VaR}(A) + \mathrm{VaR}(B)$, meaning VaR can penalize diversification.

3.3 Expected Shortfall (CVaR)

Definition 3.2 (Expected Shortfall)**.**

$$\mathrm{ES}_\alpha(L) = \frac{1}{1-\alpha} \int_\alpha^1 \mathrm{VaR}_u(L)\, du$$

When F_L is continuous, this simplifies to:

$$\mathrm{ES}_\alpha(L) = \mathbb{E}[L \mid L > \mathrm{VaR}_\alpha(L)]$$

ES is the *average loss in the tail* beyond VaR. It directly answers: "When things go wrong, how wrong do they go on average?"

Example 3.3 (ES vs. VaR)**.** For the same SPY/TLT portfolio, if $\text{VaR}_{0.95} = \$1{,}850$, then $\text{ES}_{0.95}$ averages the 13 worst losses. Suppose those are \$1,850, \$2,100, \$2,300, ..., \$8,200. Their average might be \$3,400. So $\text{ES}_{0.95} = \$3{,}400$ — nearly double the VaR.

3.4 Coherent Risk Measures

Artzner et al. (1999) defined four axioms a "sensible" risk measure ρ should satisfy:

1. **Monotonicity**: If $L_1 \leq L_2$ always, then $\rho(L_1) \leq \rho(L_2)$.
2. **Subadditivity**: $\rho(L_1 + L_2) \leq \rho(L_1) + \rho(L_2)$. Diversification does not increase risk.
3. **Positive homogeneity**: $\rho(\lambda L) = \lambda\,\rho(L)$ for $\lambda > 0$.
4. **Translation invariance**: $\rho(L + c) = \rho(L) + c$. Adding cash reduces risk dollar-for-dollar.

Key Insight

ES is coherent. VaR is not (it fails subadditivity). This is the theoretical reason Basel III/IV replaced VaR with ES for market risk capital: the Fundamental Review of the Trading Book (FRTB) mandates $\text{ES}_{0.975}$ instead of $\text{VaR}_{0.99}$.

Go Deeper

- Artzner, Delbaen, Eber & Heath (1999). "Coherent measures of risk." *Mathematical Finance*, 9(3), 203–228. The foundational paper.
- Acerbi & Tasche (2002). "On the coherence of expected shortfall." *Journal of Banking & Finance*, 26(7), 1487–1503.
- Jorion (2007). *Value at Risk*. The practitioner's reference.

4 Extreme Value Theory: Modeling the Worst Cases

Standard distributional fits (normal, t) apply to the *bulk* of the distribution. For the extreme tails — the 1% or 0.1% worst outcomes — Extreme Value Theory (EVT) provides a rigorous framework.

4.1 The Key Theorem

Theorem 4.1 (Pickands–Balkema–de Haan)**.** *For a wide class of distributions, the excess loss above a high threshold u converges in distribution to the* ***Generalized Pareto Distribution*** *(GPD):*

$$\mathbb{P}(L - u \leq y \mid L > u) \approx G_{\xi,\beta}(y) = \begin{cases} 1 - (1 + \xi y/\beta)^{-1/\xi} & \xi \neq 0 \\ 1 - e^{-y/\beta} & \xi = 0 \end{cases}$$

where ξ is the ***shape*** *(tail index) and $\beta > 0$ is the scale.*

Example 4.1 (Tail index interpretation)**.**

- $\xi > 0$ (heavy tail): Pareto-like. Typical for equity returns ($\xi \approx 0.2$–0.4).
- $\xi = 0$ (light tail): exponential decay. Rare in finance.

- $\xi < 0$ (bounded tail): finite upper endpoint. Insurance losses sometimes.

Fitting GPD to S&P daily losses above the 95th percentile typically gives $\hat{\xi} \approx 0.25$, confirming heavy tails and allowing extrapolation to extreme quantiles (99.9%) that have few or no historical observations.

Go Deeper

- de Haan & Ferreira (2006). *Extreme Value Theory: An Introduction.* The mathematical reference.
- McNeil & Frey (2000). "Estimation of tail-related risk measures for heteroscedastic financial time series." *Journal of Empirical Finance.* The GARCH + EVT approach.
- Embrechts, Klüppelberg & Mikosch (1997). *Modelling Extremal Events.* Classic for insurance/finance.

5 Dependence: Beyond Correlation

5.1 Why Correlation Is Not Enough

Pearson correlation ρ measures *linear* association. Two variables can have $\rho = 0$ yet be strongly dependent (e.g., $Y = X^2$). More critically for risk: assets that appear weakly correlated in normal markets can become highly correlated during crises.

Example 5.1 (Correlation breakdown)**.** From 2005–2007, the correlation between US equities and mortgage-backed securities was ≈ 0.3. During the 2008 crisis, it spiked above 0.8. A portfolio diversified under normal-market correlation was far less diversified when it mattered most.

5.2 Copulas: The Right Tool

A **copula** $C(u_1, \ldots, u_d)$ is a function that binds marginal distributions into a joint distribution. By Sklar's theorem, *any* joint distribution can be decomposed as:

$$F(x_1, \ldots, x_d) = C\big(F_1(x_1), \ldots, F_d(x_d)\big)$$

This separates two modeling choices:

- **Marginals** F_i: model each asset's return distribution individually (e.g., GARCH with skewed-t).
- **Dependence structure** C: model how they move together, especially in the tails.

Key copula families and their tail behavior:

Copula	**Lower tail dep.** λ_L	**Upper tail dep.** λ_U
Gaussian	0	0
Student-t	> 0 (symmetric)	> 0 (symmetric)
Clayton	> 0	0
Gumbel	0	> 0
Frank	0	0

Table 1: Tail dependence by copula family. For risk, lower tail dependence (joint crashes) matters most. The Gaussian copula has *none* — this was a core flaw in pre-2008 CDO pricing.

Key Insight

The Gaussian copula assigns probability zero to joint extreme co-movements. This made it catastrophically wrong for pricing CDOs and estimating portfolio credit risk before 2008. Modern practice uses t-copulas or Clayton copulas that capture tail dependence.

Go Deeper

- Nelsen (2006). *An Introduction to Copulas*. 2nd ed. The textbook.
- Joe (2014). *Dependence Modeling with Copulas*. Advanced, covers vine copulas.
- Embrechts, McNeil & Straumann (2002). "Correlation and dependence in risk management." *Risk Magazine*. Why correlation fails.

6 The Regulatory Landscape: Why This Matters

Risk models do not exist in a vacuum. Banks, insurers, and asset managers operate under regulatory frameworks that dictate *which* risk measures to compute, *how* to compute them, and *how much capital* to hold.

6.1 Basel Framework (Banking)

Accord	Era	Key change for risk measurement
Basel I (1988)	Simple	Flat capital charges by asset class. No internal models.
Basel II (2004)	VaR era	Banks allowed to use internal VaR models (99%, 10-day).
Basel III (2010)	Post-crisis	Stressed VaR, capital buffers, leverage ratio added.
Basel III.1 / "IV" (2017–2028)	Current	**FRTB**: replaces VaR with ES at 97.5%. Output floors on internal models. Standardized Measurement Approach for operational risk.

FRTB (Fundamental Review of the Trading Book):

- Market risk capital: $\mathrm{ES}_{0.975}$ over a base horizon of 10 days, computed with different liquidity horizons per risk factor.
- Internal Models Approach (IMA): banks must pass backtesting at the desk level.
- Standardized Approach (SA): sensitivities-based method (delta, vega, curvature) for banks without IMA approval.
- Implementation: EU phasing 2025–2026, UK targeted 2027, US "Basel Endgame" from July 2025.

6.2 What This Means for You

Every project in this repo connects to real regulatory or industry requirements:

- **P01** (Dashboard): implements VaR and ES with Basel-mandated backtesting.

- **P02** (Monte Carlo): the computational engine behind IMA.
- **P03** (Credit Scoring): PD models under Basel IRB, with SR 11-7 model governance.
- **P04** (Volatility): GARCH models feed into ES and VaR calculations.

Go Deeper

- Basel Committee (2019). "Minimum capital requirements for market risk." BCBS d457. The FRTB standard.
- Federal Reserve (2011). SR 11-7: "Guidance on Model Risk Management." Gold standard for model governance.
- Hull (2018). *Risk Management and Financial Institutions*. Ch. 15–17 on Basel.

7 Stochastic Processes for Risk

You already know stochastic processes from your actuarial training. Here we focus on the specific models used in financial risk and *why* each matters.

7.1 Geometric Brownian Motion (GBM)

The workhorse of Monte Carlo simulation:

$$dS_t = \mu S_t\,dt + \sigma S_t\,dW_t \quad \Longrightarrow \quad S_T = S_0 \exp\Big[\Big(\mu - \tfrac{\sigma^2}{2}\Big)T + \sigma W_T\Big]$$

Used in: option pricing (Black-Scholes), portfolio simulation, VaR by Monte Carlo.

Limitation: constant volatility σ. Markets exhibit volatility clustering — addressed by GARCH (discrete) and stochastic volatility models (continuous).

7.2 GARCH(1,1): Volatility That Remembers

$$r_t = \mu + \epsilon_t, \quad \epsilon_t = \sigma_t z_t, \quad z_t \sim D(0,1)$$
$$\sigma_t^2 = \omega + \alpha\epsilon_{t-1}^2 + \beta\sigma_{t-1}^2$$

where D is typically Student-t or skewed-t.

Example 7.1 (GARCH intuition)**.** If yesterday's return was large ($|\epsilon_{t-1}|$ big), today's conditional variance σ_t^2 increases. The β term gives persistence: high volatility regimes last for weeks or months. For the S&P 500, typical estimates are $\alpha \approx 0.09$, $\beta \approx 0.90$, giving persistence $\alpha + \beta \approx 0.99$ — volatility shocks decay very slowly.

Extensions (covered in Project 04): GJR-GARCH (leverage effect: negative returns increase volatility more), EGARCH (log-volatility, no positivity constraint), APARCH (power transformations).

7.3 Mean-Reverting Processes

For interest rates and credit spreads, the Ornstein-Uhlenbeck / Vasicek model:

$$dr_t = \kappa(\theta - r_t)\,dt + \sigma\,dW_t$$

Mean-reverts to θ at speed κ. Used in: yield curve modeling, CVA calculations (Project 09), credit intensity models.

Go Deeper

- Glasserman (2003). *Monte Carlo Methods in Financial Engineering.* Ch. 3 (GBM simulation), Ch. 8 (variance reduction).
- Engle (1982). "Autoregressive conditional heteroscedasticity." *Econometrica.* The ARCH origin.
- Bollerslev (1986). "Generalized autoregressive conditional heteroscedasticity." *Journal of Econometrics.* GARCH(1,1).

8 Summary: The Mental Model

Layer	**What it does**	**Key tools**
Marginals	Model each risk factor's distribution	GARCH, Student-t, skewed-t, EVT (GPD)
Dependence	Model how risk factors move together	Copulas (Gaussian, t, Clayton, vine)
Aggregation	Combine into portfolio loss distribution	Monte Carlo, variance reduction, FFT
Risk measures	Summarize the loss distribution	VaR, ES, spectral measures
Validation	Check if the model works	Backtesting (Kupiec, Christoffersen), stress testing

Every project in the curriculum operates within this pipeline. The first four projects build it from the ground up: the dashboard (P01) computes measures and validates them; the MC engine (P02) handles aggregation; credit scoring (P03) models a different kind of marginal (default probability); and volatility modeling (P04) improves the marginal layer.